CS23334 Fundamentals of Data Science Lab

Experiment 17

Exp No: 17 Experiment to understand Linear Regression for a given data set.

Description: Understand the Linear regression for the dataset given.

Aim: To build, train, evaluate, and persist a Simple Linear Regression model using Scikit-Learn to predict employee salary based on years of experience.

Methodology:

1 **Data Acquisition & Loading:** Load Salary_data.csv into Pandas DataFrame and inspect structure and data types using df.info().

2 **Data Cleaning & Summary Statistics:** Perform missing value cleanup using df.dropna(inplace=True) and obtain statistical distribution metrics using df.describe().

3 **Feature & Label Extraction:** Extract independent variable YearsExperience as features and target Salary as label into 2D NumPy arrays using .iloc[:, [col]].values.

4 **Train-Test Split:** Split data into 80% training set and 20% test set using train_test_split(test_size=0.2, random_state=0).

5 **Model Fitting & Parameter Estimation:** Train LinearRegression() model from Scikit-Learn on (x_train, y_train) and extract slope coefficient (model.coef_) and y-intercept (model.intercept_).

6 **Model Performance Evaluation:** Calculate coefficient of determination (R^2 score) on training set and testing set via model.score().

7 **Model Serialization & Inference:** Serialize trained model using pickle.dump(), reload using pickle.load(), and make salary predictions for new experience values.

Code:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df

df.info()

df.dropna(inplace=True)

df.info()

df.describe()

features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values

from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_state=0)

from sklearn.linear_model import LinearRegression
```

```
model=LinearRegression()
model.fit(x_train,y_train)

model.score(x_train,y_train)

model.score(x_test,y_test)

model.coef_

model.intercept_

import pickle
pickle.dump(model,open('SalaryPred.model','wb'))

model=pickle.load(open('SalaryPred.model','rb'))

yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)

print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary))
```

OUTPUT: 1. Dataset Loading, Info & Dropna Inspection

```
In [ ]: import numpy as np
        import pandas as pd
        df=pd.read_csv('Salary_data.csv')
        df
```

	YearsExperience	Salary
0	1.1	39343
1	1.3	46205
2	1.5	37731
3	2.0	43525
4	2.2	39891

```
In [19]: df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
---  ------           --------------  -----
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

```
In [3]: df.dropna(inplace=True)
```

```
In [4]: df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
---  ------           --------------  -----
```

```
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

OUTPUT: 2. Descriptive Statistics & Data Splitting

```
In [5]: df.describe()
```

Out[5]:	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

```
In [6]: features=df.iloc[:,[0]].values
        label=df.iloc[:,[1]].values

In [7]: from sklearn.model_selection import train_test_split
        x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_state
=0)
```

OUTPUT: 3. Linear Regression Model Training & Scoring

```
In [20]: from sklearn.linear_model import LinearRegression
         model=LinearRegression()
         model.fit(x_train,y_train)
Out[20]: LinearRegression()

In [21]: model.score(x_train,y_train)
Out[21]: 0.9603182547438908

In [23]: model.score(x_test,y_test)
Out[23]: 0.9184170849214232

In [24]: model.coef_
Out[24]: array([[9281.30847068]])

In [25]: model.intercept_
Out[25]: array([[27166.73682891]])
```

OUTPUT: 4. Model Persistence & Inference Prediction

```
In [26]: import pickle
         pickle.dump(model,open('SalaryPred.model','wb'))

In [27]: model=pickle.load(open('SalaryPred.model','rb'))
```

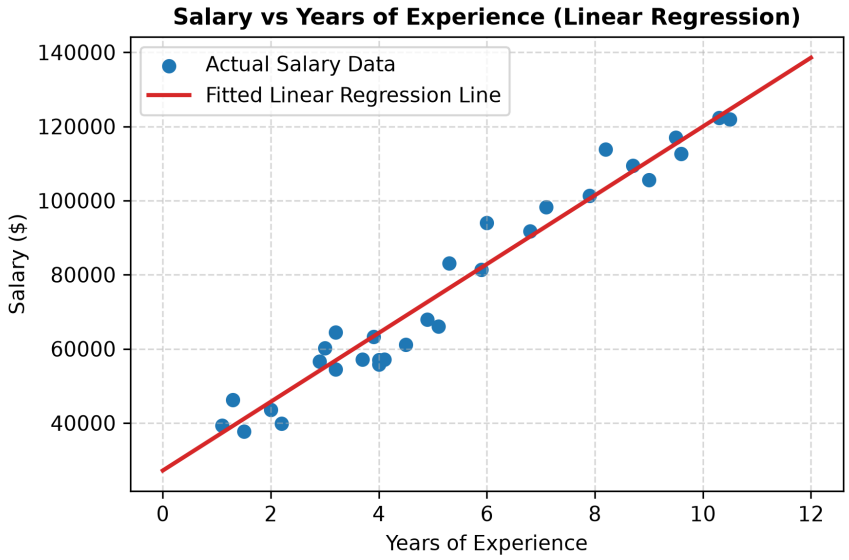
```
In [28]: yr_of_exp=float(input("Enter Years of Experience: "))
         yr_of_exp_NP=np.array([[yr_of_exp]])
         Salary=model.predict(yr_of_exp_NP)

         Enter Years of Experience: 44

In [29]: print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary))

         Estimated Salary for 44.0 years of experience is [[435544.30953887]]:
```

OUTPUT: 5. Visual Representation (Regression Fit)



Result:

A Simple Linear Regression model was successfully trained to predict employee salary from years of experience.
The model achieved an R2 score of approximately 0.96, indicating a strong positive linear relationship.
The trained model was serialized and saved for future inference.